

Thalamocortical coherence predicts persistent postconcussive symptoms

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ABSTRACT

The pathogenetic mechanism of persistent post-concussive symptoms (PCS) following concussion remains unclear. Thalamic damage is known to play a role in PCS prolongation while the evidence and biomarkers that trigger persistent PCS have never been elucidated. We collected longitudinal neuroimaging and behavior data from patients and rodents after concussion, complemented with rodents' histological staining data, to unravel the early biomarkers of persistent PCS. Diffusion tensor imaging (DTI) were acquired to investigate the thalamic damage, while quantitative thalamocortical coherence was derived through resting-state functional MRI for evaluating thalamocortical functioning and predicting long-term behavioral outcome. Patients with prolonged symptoms showed abnormal DTI-derived indices at the boundaries of bilateral thalami (peri-thalamic regions). Both patients and rats with persistent symptoms demonstrated enhanced thalamocortical coherence between different thalamocortical circuits, which disrupted thalamocortical multifunctionality. In rodents, the persistent DTI abnormalities were validated in thalamic reticular nucleus (TRN) through immunohistochemistry, and correlated with enhanced thalamocortical coherence. Strong predictive power of these coherence biomarkers for long-term PCS was also validated using another patient cohort. Postconcussive events may begin with persistent TRN injury, followed by disrupted thalamocortical coherence and prolonged PCS. Functional MRI-based coherence measures can be surrogate biomarkers for early prediction of long-term PCS.

Summary

Persistent postconcussive symptoms are associated with long-lasting

thalamic reticular nucleus injury and disrupted thalamocortical coherence. Functional MRI-based coherence biomarkers can be considered surrogate biomarkers for early prediction of long-term postconcussive

Abbreviations: BAI, Beck Anxiety Inventory; BDI, Beck Depression Inventory; CHI, closed-head injury; DTI, diffusion tensor imaging; FA, fractional anisotropy; FDR, False Discovery Rate; fMRI, functional MRI; GOSE, Glasgow Outcome Scale-Extended; HC, healthy control; MMSE, Mini-Mental State Examination; mTBI, mild traumatic brain injury; PCSQ, Rivermead Post Concussion Symptoms Questionnaire; PCS, postconcussive symptoms; PSQI, Pittsburgh sleep quality index; RD, radial diffusivity; TCD, thalamocortical dysrhythmia; TD, track density; TRN, thalamic reticular nucleus.

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symptoms.

1. Introduction

Concussion, a mild traumatic brain injury (mTBI) caused by a blow to the head that make the brain to move and hit the skull, typically present with neurological dysfunctions such as headache, sleep disturbances, depression, and cognitive impairment, which typically resolve within a few months (Vanderploeg et al., 2009; Veltman et al., 1996). However, a subset of patients may have persistent postconcussive symptoms (PCS) over years (Alves et al., 1993; Chen et al., 2022; Corrigan et al., 2013), but the pathogenetic mechanism remained poorly understood. There is also growing concerns regarding the potential long-term sequelae of concussion including chronic neurologic issues related to stepwise neurocognitive impairment in later life (Chen et al., 2022; Corrigan et al., 2013). Hence, clarifying the pathogenetic mechanisms underlying various persistent PCS is essential to identify early biomarkers of prolonged PCS.

On the basis of a simulation model of shearing stress distribution within the brain following concussive injury, studies have proposed that rotational acceleration produces maximum shearing forces mainly distributed in the center of the brain (Mendez et al., 2005; Ropper and Gorson, 2007), also known as the peri-thalamic region. These forces have been reported to cause thalamocortical disruptions, which a human study show could last longer even than cortical dysfunction (Moeller et al., 2011). The thalamus is a centrally located relay station with nerve fibers projecting out to the cerebral cortex in all directions for information transmission. Thalamocortical interactions include both temporal synchronization between specific thalamic subnuclei and corresponding cortical regions and desynchronization between irrelevant areas (Maxwell et al., 2004). If such interactions are out of sync, multiple neuropsychiatric symptoms, such as neuropathic pain (Jeanmonod et al., 2001), tinnitus (De Ridder et al., 2015), Parkinson's disease (Magnin et al., 2001), schizophrenia (Schulman et al., 2011), epilepsy (Martire et al., 2019), and depression (Ben-Simon et al., 2008), may arise. In addition, thalamus has also been implicated to be related to the development and recovery of PCS from prior studies (Anderson et al., 1996; Mofakham et al., 2022). With its global long-range connections, thalamus plays a role in the neuroplasticity mechanism in the injured brain and can facilitate the improvement of PCS (Munivenkataappa and Agrawal, 2016). One previous study has also reported that patients with multi-year coma after traumatic brain injury exhibited significant improvement after thalamic stimulation (Schiff et al., 2007). In other words, the thalamic injuries may potentially lead to the failure in postconcussive neuroplasticity mechanism, resulting in the prolonged PCS. Despite the importance of thalamocortical interaction, and the negative impacts of thalamic injury on consciousness and cognition, little is known about the mechanism that sets off such chain of events that consequently lead to multiple neurological and psychiatric symptoms.

Since the thalamus contributes to a broad range of critical functions, the side effects of thalamic and/or thalamocortical damage can vary from person to person. We first hypothesized that 1) the extent of concussion-induced thalamic damage and the disturbed thalamocortical coherence should be quantifiable, and be able to explain not only clinical between-group differences (e.g., patient vs. healthy control [HC]), but also within-group variances (e.g., symptom-prolonged patients versus symptom-resolved patients); 2) as thalamocortical coherence is restored, so should one's PCS. To this end, we longitudinally recruited 70 patients with concussion and 70 demographically-matched healthy controls (HCs), and followed the patients for 2 years on their neuroimaging and neuropsychological data.

First, we aimed to test whether peri-thalamic injury could be observed after concussion and which one of the thalamic subdivision is most sensitive to concussion-induced shearing forces. These peri-thalamic microstructural injuries were assessed using diffusion tensor

imaging (DTI). Because the histopathology, which is considered to be the gold standard for assessing the pathophysiological changes following concussion, can be difficult to obtain since patients do not die from such mild injury, we also constructed a translational rodent closed-head injury (CHI) model to mimic the clinical status of our patients after different degrees of concussion-induced shearing forces. Then, the histological staining would be applied to the sacrificed rodents to further validate the pathological changes assessed by DTI, especially at the perithalamic regions.

Then, to investigate the dynamic configurations of thalamocortical interactions in normal and different levels of disease states, here we used functional MRI (fMRI) to dynamically record the brain activity of both humans and rodents during resting state. Specifically, we investigated the mechanism of thalamocortical functional connections at both thalamic and cortical levels, and tested whether the extent of perithalamic damage can explain the severity of thalamocortical disruption. Functional connectivity was recently used to investigate disrupted thalamocortical functional interactions (Banks et al., 2016; Sours et al., 2015; Zhou et al., 2014). However, reported results are inconsistent, with some groups demonstrating a decrease in thalamocortical functional connectivity after concussion (Banks et al., 2016; Kim et al., 2022), and others an increase (Sours et al., 2015; Zhou et al., 2014). One of the key contributors to these conflicting results may be the inaccurate measurements of thalamocortical connections due to the delayed interregional signaling, particularly in patients after mTBI. Functional connectivity estimated through simple correlational analyses with zero-time lag is extremely sensitive to the interregional delays (Li et al., 2021; Sun et al., 2004); this connectivity may be more pronounced when inefficient signal transmission is caused by concussion-induced axonal damage, myelin damage, or both. In this study, we proposed to use temporal coherence (Fig E1 [online]) a delay-independent estimate (Sun et al., 2004), to investigate concussion-induced thalamocortical disruptions. Then, we determined whether the disturbed thalamocortical coherence is associated with PCS prolongation by correlating the quantitative thalamocortical coherence measures assessed at baseline with the long-term PCS estimates. If so, changes in thalamocortical coherence can possibly yield reliable biomarkers that are helpful for early prediction of long-term PCS after concussion. Finally, a machine learning-based predictive model was trained using the baseline quantitative thalamocortical coherence measures and long-term PCS estimates, and validated with external cohort data to demonstrate its predictive power for persistent PCS.

2. Methods

2.1. Participants and neuropsychological testing

Patients with concussion (defined by the following criteria: closed-head injury (CHI), loss of consciousness lasting < 30 min, initial Glasgow Coma Scale score > 13, and normal head CT results) and demographically matched healthy controls (HCs) were recruited from Taipei Medical University Hospital (TMUH) between September 2015 and September 2021 (Table 1). Their baseline assessments were collected within the first two weeks of injury, and follow-ups were conducted at 1 and 2 years after mTBI. This study was approved by the Joint Institutional Review Board of TMUH (Nos. 201504083, N201612008, N201904032, and N202102008).

The exclusion criteria were any history of prior brain injury, coexisting or previous neurological or psychiatric illness, and contraindication for MRI. Exclusion criteria for the HCs were the same as the additional requirement of no history of concussion or brain injury.

Glasgow Outcome Scale-Extended (GOSE) and Mini-Mental State Examination (MMSE) were administered by a clinical psychologist during baseline assessment. Neuropsychological assessments, including Pittsburgh sleep quality index (PSQI), Rivermead Post Concussion Symptoms Questionnaire (PCSQ), Beck Anxiety Inventory (BAI), and

Table 1

Demographic, Clinical, and Cognitive Characteristics and Working Memory Task Performance of Enrolled Individuals.

Demographic Characteristics	HCs	Patients With mTBI (Baseline)	Patients With mTBI (1-y Follow-Up)	Patients With mTBI (2-y Follow-Up)
Sample size (n)	70	70	30	17
Sex (female/male)	47/23	47/23	22/8	9/8
Age (y)	36.66 ± 12.04	37.97 ± 12.24	36.80 ± 11.29	35.76 ± 10.97
Education (y)	15.54 ± 3.98	14.45 ± 3.03	14.65 ± 2.76	14.71 ± 2.82
GOSE score	N.A.	6.20 ± 1.98	6.10 ± 2.11	5.63 ± 2.16
MMSE score	29.47 ± 0.74	28.44 ± 1.61 ***	28.80 ± 1.45*	28.47 ± 1.59**
Postinjury days (initial scan; d)	N.A.	6.76 ± 2.85	5.83 ± 3.15	5.29 ± 2.69
Postinjury days (follow-up; d)	N.A.	N.A.	373.83 ± 14.60	739.59 ± 17.08
Clinical Characteristics				
PSQI	4.86 ± 2.20	8.07 ± 4.36***	6.67 ± 9.45**	7.35 ± 4.05**
PCSQ score	4.94 ± 5.22	13.30 ± 13.43***	9.03 ± 9.45*	9.29 ± 11.66*
BAI score	3.91 ± 4.32	9.80 ± 9.57***	7.27 ± 7.84*	7.43 ± 8.06
BDI score	4.56 ± 4.61	9.70 ± 9.86**	6.97 ± 6.18*	7.93 ± 6.57*

Note.—Demographic and behavioral characteristics compared between patients with mTBI and HCs (* $P < .05$, ** $P < .01$, *** $P < .001$). Data presented as means ± standard deviations.

The MMSE score, PSQI, PCSQ score, BAI score, and BDI score differed significantly between patients with mTBI and HCs. These symptoms did not improve significantly at the 1-year follow-up. The PSQI, PCSQ score, and BDI score did not improve significantly, even after 2 years of mTBI.

BAI = Beck Anxiety Inventory; BDI = Beck Depression Inventory; GOSE = Glasgow Outcome Scale-Extended; HC = healthy control; MMSE = Mini-Mental State Examination; mTBI = mild traumatic brain injury; PCSQ = Rivermead Post Concussion Symptoms Questionnaire; PSQI = Pittsburgh sleep quality index.

Beck Depression Inventory (BDI), were administered on the same day as the initial and follow-up scans.

2.2. Rodent resource and research protocol

Twenty-four adult (8-week-old) male Sprague Dawley (SD) rats weighing 250–400 g were obtained from BioLASCO Taiwan (Taipei, Taiwan). They were randomly assigned to three groups (single CHI, double CHI, and sham control) for longitudinal and cross-sectional assessment (Fig E2 [online]). In each group, one rat was sacrificed 24 h after CHI for histological staining, whereas the others were administered MRI and behavioral assessments at the designated time points to dynamically monitor post-CHI behavioral and pathological changes; they were sacrificed after 7 weeks for long-term histological staining. Note that the rats sacrificed at 24 h after CHI were treated as test animals and not included in further comparison.

The animal experiments were performed in accordance with the recommendations of National Institutes of Health Guidelines for Animal Research's *Guide for the Care and Use of Laboratory Animals* and the *Animal Research: Reporting in vivo Experiments Guidelines*. The research protocol was approved by the Institute Animal Care and Use Committee at Taipei Medical University (LAC-2019-0358 and LAC-2020-0437). The details of the rodent research protocol are described in Appendix E1 [online].

2.3. Human MRI data acquisition and experimental design

MRI data were obtained using a 3 T MR scanner (Siemens MAGNETOM Prisma, Erlangen, Germany) equipped with a 20-channel head coil.

Diffusion tensor imaging (DTI) was performed through echo-planar imaging (EPI) with two b values (1000 and 2500 s/mm²) along 64 noncollinear and noncoplanar diffusion directions and 10 b_0 images. The imaging parameters were TR/TE = 7500/59 ms, flip angle = 90°, acquisition matrix = 256 × 256 × 50 with resolution = 0.86 × 0.86 × 3 mm². Standard single-shot gradient-echo EPI-based fMRI (TR/TE = 2000/20 ms, flip angle = 90°, voxel size = 3 × 3 × 3.5 mm³, matrix = 64 × 64 × 40, 190 volumes) was also performed. Participants were requested to keep their eyes closed and not engage in any thoughts while remaining awake, alert, and as motionless as possible (Li et al., 2021). Specifically, participants were advised to avoid consuming caffeine, alcohol, or other stimulants, such as energy drinks, tea, or coffee, before the scan to ensure that they were well-rested and alert (Kawagoe et al., 2018; Lee et al., 2013). Additionally, they were advised to get adequate rest and avoid intense physical activity before the scan to avoid fatigue or drowsiness that might interfere with the resting-state measurement (Lee et al., 2013; Smitha et al., 2017). We also provided participants with practice runs before the actual scan to help them become familiar with the procedures and reduce anxiety or discomfort during the actual scan (Kawagoe et al., 2018).

For the coregistration and normalization of fMRI and DTI data, three-dimensional T₁-weighted magnetization-prepared rapid gradient-echo images (MP-RAGE; TR/TE/TI = 2300/3.26/1030 ms, flip angle = 8°, voxel size = 1 × 1 × 1 mm³, matrix = 256 × 256 × 176) were obtained.

2.4. DTI analysis

DTI data were preprocessed using FSL 5.0.10 (FMRIB, Oxford, UK) and MRtrix 3.0.2 (Brain Research Institute, Melbourne, Australia). Specifically, denoising, Gibbs ringing artifact removal, and bias field correction were applied using MRtrix and then processed according to the standard FSL pipeline (ie, brain mask estimation along with eddy-current and motion correction). Fixel-based analysis of fiber density was performed using the MRtrix multishell and multitissue constrained spherical deconvolution method (Dhollander and Connelly, 2016) and the probabilistic streamlining method (Tournier et al., 2010). DTI-derived parametric maps, including fractional anisotropy (FA), radial diffusivity (RD), and track density (TD) maps, were generated and then spatially normalized to the template space (for humans: Montreal Neurological Institute template [MNI-152]; for rats: Paxinos and Watson Stereotaxic Coordinates (Paxinos and Watson, 2006; Valdes-Hernandez et al., 2011)) by using MRtrix.

2.5. fMRI preprocessing

The fMRI data were preprocessed using Statistical Parametric Mapping (SPM12; Wellcome Department, University College London, UK) for slice timing correction, realignment, spatial normalization to template space, and spatial smoothing with a Gaussian kernel (for humans: 5-mm full-width-at-half-maximum; for rats: 0.5-mm full-width-at-half-maximum). Linear and quadratic trends of the fMRI time series were also removed.

2.6. Atlas-based functional connectivity and functional coherence computation

The human cortical regions of interest were defined using Brodmann area (BA) (Brodmann, 1909) at the lobar level and the thalamic sub-nuclei were delineated using the Talairach atlas (Lancaster et al., 2000). The six head-motion parameters and averaged white matter and cerebrospinal fluid signals were partially regressed out of the preprocessed fMRI time series. The within-thalamic functional connectivity was estimated by calculating the Pearson correlation coefficients between the temporal filtered (0.01–0.08 Hz) fMRI time series (Biswal et al., 1995; Li et al., 2021), followed by Fisher Z transform. The magnitude-squared thalamocortical functional coherence (Sun et al., 2004) was calculated

using in-house MATLAB (version R2020a; Mathworks, Sherborn, MA, USA) scripts. Note that the frequency band selection in fMRI functional connectivity and coherence analysis can have a significant impact on the results, making the determination of the disease-dominant frequency range crucial for further analysis of the connectome matrix. However, the highly idiosyncratic nature of concussion led to differences in each patient's collision angle, position, and structural damage, resulting in variations in thalamocortical injury between individuals. Selecting the specific thalamic subnuclei or cortical region for disease-dominant frequency range estimation can, therefore, be challenging. To overcome this issue and determine the disease-dominant frequency range that most accurately reflected the disturbed thalamocortical coherence, we initially performed magnitude-squared coherence analysis between the averaged time series of the voxels from the entire thalami and each cortical region in the frequency range of 0.01–0.25 Hz (Zhou et al., 2014). Subsequently, the frequency range exhibiting significant difference in coherence values between patients and HCs was regarded as the dominant frequency range and utilized for further analysis.

2.7. fMRI-Based functional connectome matrix and connectome biomarkers

Two connectome matrices were generated: the within-thalamic functional connectivity matrix and the thalamocortical functional coherence matrix. Specifically, the within-thalamic functional connectivity matrix was estimated from the temporal filtered fMRI time series of each pair of thalamic subnuclei in the disease-dominant frequency range. Similarly, the thalamocortical functional coherence matrix was computed between each thalamic subnucleus and BA cortical region pair. In other word, each element in the connectome matrix represents the functional connectivity/coherence value estimated between the specific region-of-interest (ROI) pairs (Amico and Goñi, 2018; Smith et al., 2013; Turk et al., 2019).

Due to the highly idiosyncratic nature of concussion, which leads to differences in each patient's collision angle, position, and structural damage, the concussion-induced thalamic and/or thalamocortical damage can vary from person to person. Therefore, to quantify the level of changes in within-thalamic functional connectivity and yield a potential personalized disease burden of thalamic dysfunction, the within-thalamic functional connectome matrix was then simply averaged at the individual level as a compromise strategy. Similarly, the thalamocortical coherence matrix was also averaged at the individual level as a compromise strategy to quantify the degree of pathologically altered thalamocortical coherence. The averaged within-thalamic functional connectivity and the averaged thalamocortical coherence measure were treated as the connectome biomarkers.

2.8. Statistical analyses

A one-sample *t* test was employed to determine the within-group significance in the HC and mTBI groups, whereas a two-tailed two-sample *t* test was used to determine between-group differences. A two-tailed paired-sample *t* test was applied to examine the significant differences between the initial and follow-up data. Pearson correlation coefficients were computed to reveal the relationships between functional coherence and neuropsychological or behavioral changes. Moreover, we evaluated the linear relationship between the proposed connectome biomarkers and neuropsychological or behavioral outcomes using a generalized linear model (GLM), while controlling for confounds such as age and sex. The statistical tests were all corrected by controlling the false discovery rate (FDR) at $q = 0.01$ to prevent errors related to multiple comparisons.

2.9. Individualized prediction of the prolonged PCS by connectome biomarkers

The connectome biomarkers obtained through baseline measurement, such as the averaged within-thalamic functional connectivity and the averaged thalamocortical coherence, along with patient's age and sex were treated as the possible features to train the support vector machine regressor with leave-one-out cross-validation on each prediction. The machine learning-based predictive model was applied using in-house MATLAB (version R2020a, Mathworks, Sherborn, MA, USA) scripts.

3. Results

3.1. Demographic characteristics

In total, 70 patients with mTBI (age = 37.9 ± 12.2 y; 47 [67.1%] female) and 70 demographically matched HCs (age = 36.7 ± 12.0 y; 47 [67.1%] female) were recruited (Fig. 1). The follow-up rates at 1 and 2 years for the patients were 42.9% ($n = 30$) and 24.3% ($n = 17$), respectively (Table 1); 19 and 10 of all patients had persistent symptoms at the 1- and 2-year follow-up (Table 2).

3.2. Structural injuries after concussion

3.2.1. Clinical human results

Fig. 2 demonstrates thalamocortical tracts injury and peri-thalamic injury centered around the thalamic reticular nucleus (TRN) on DTI. At the boundaries of the bilateral thalami, where thalamic reticular nucleus (TRN) was located, we observed a significant increase in radial diffusivity (RD) and decrease in fractional anisotropy (FA) in patients compared with HCs (Table 3; Fig. 2A: yellow arrows) at baseline as well as at 1- and 2-year follow-ups. These may indicate long-lasting axonal damage, myelin damage, or both in the peri-thalamic TRN after concussion. Furthermore, a significant decrease was noted in the white matter tract density (TD) of patients (Fig. 2B) at baseline, suggesting the presence of white matter tract damage. This tract damage was resolved within 1 year after concussion, indicating that tract damage is temporary and reversible compared with the long-lasting TRN damage.

3.2.2. Translational rodent results

Similar to the clinical results of our patients, changes in RD, FA, and TD were noted in the bilateral TRN of in our rats at 1-week follow-up (Fig E3A-C [online]), which sustained for up to 7 weeks (Fig E3D-F [online]) after CHI, suggesting a long-lasting axonal and/or myelin injury in TRN after concussion. We also revealed different degrees of TRN damage after receiving single or double CHI (Fig E3 [online]). By contrast, the damage to the thalamocortical tract could be found within the first 3 weeks of CHI and disappeared afterwards (Fig E3G-I [online]). These results provided converging evidence for the notion that TRN injury may last longer than white matter tract injury.

Histological staining of the test rat in each group obtained 24 h after injury showed different levels of increased GFAP immunoreactivity in the cortex and TRN after single or double CHI, as observed through visual comparison (Fig E4A [online]). These results are consistent with their corresponding DTI on structural damages, indicating the different extents of reactive astrogliosis produced by the neuroinflammation process after injury (Acabchuk et al., 2016; Kao et al., 2019; Mierzwa et al., 2015; Mountney et al., 2017). The neuroinflammatory changes in the TRN sustained up to 7 weeks after CHI, whereas cortical damage were less pronounced and nonsignificant compared with the sham controls after 7 weeks, re-confirming the persistence of TRN injury over the 7 days.

Immunohistological staining for MBP performed 24 h after CHI in the test rat in each group (see Fig E4B [online]) revealed a reduction in myelin integrity along the cortical fiber tract and TRN region,

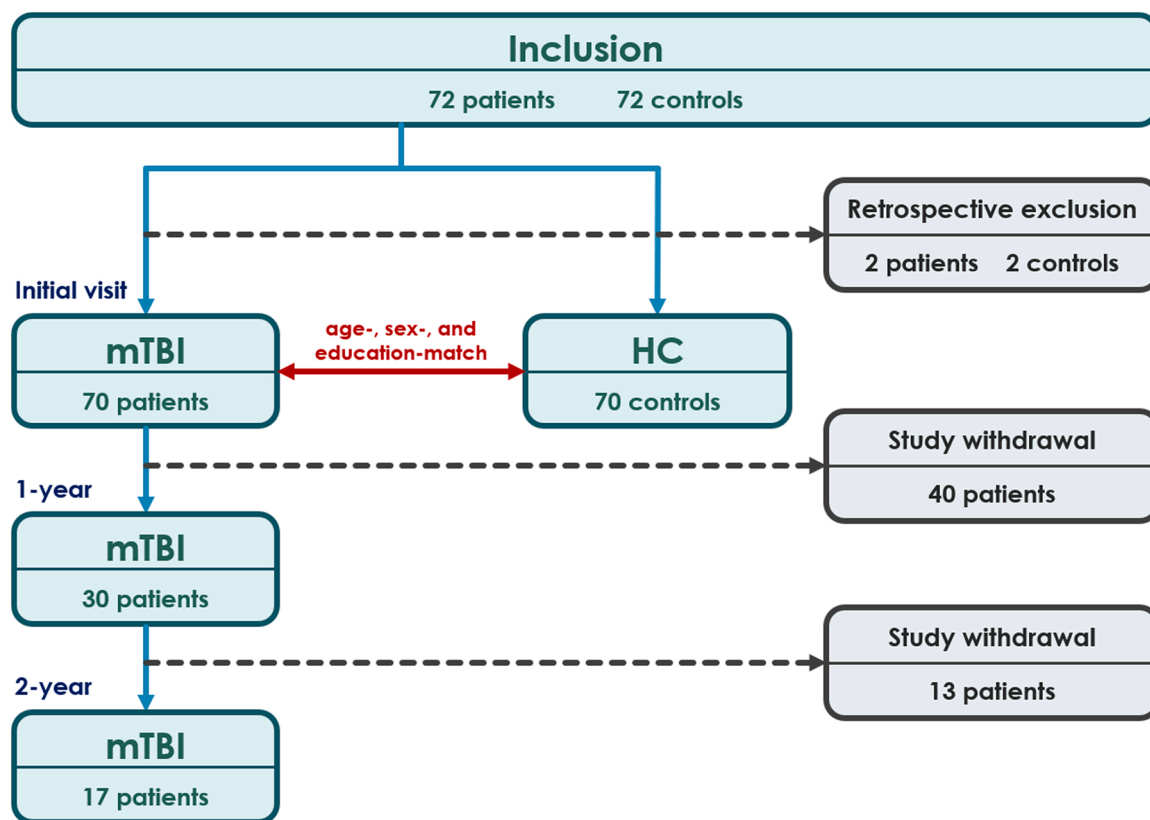


Fig. 1. Flowchart of participant recruitment. One patient and two healthy controls (HCs) were retrospectively excluded because of severe MRI artifacts due to dental prosthesis, and another patient was excluded due to pituitary gland tumor diagnosis. The injury mechanisms were as follows: motor vehicle accident ($n = 37$), fall ($n = 17$), sports ($n = 3$), assault ($n = 8$), and other ($n = 5$). The main reasons for patient dropout in the follow-up visits included unresponsiveness to phone calls or emails and change in address or job.

Table 2

Demographic, Clinical, and Cognitive Characteristics and Working Memory Task Performance of Individuals with Prolonged or Resolved Symptoms at 1- and 2-Year Follow-up.

Demographic Characteristics	HCs	Patients With mTBI at Baseline	mTBI Patients With Symptoms Resolved Within 1 y	mTBI Patients With Symptoms Prolonged Over 1 y	mTBI Patients With Symptoms Resolved Within 2 y	mTBI Patients With Symptoms Prolonged Over 2 y
Sample size (n)	70	70	11	19	7	10
Sex (female/male)	47/23	47/23	5/6	17/2 ††	1/6**††	8/2 ††
Age (y)	36.66 ± 12.04	37.97 ± 12.24	28.91 ± 7.20 * †	41.37 ± 10.79 ††	32.33 ± 10.39*†	38.60 ± 11.42
Education (y)	15.54 ± 3.98	14.45 ± 3.03	15.45 ± 1.75	14.18 ± 3.16	15.33 ± 2.34	14.20 ± 3.22
GOSE score	N.A.	6.20 ± 1.98	6.80 ± 1.81	5.74 ± 2.21	6.40 ± 2.30	5.50 ± 2.07
MMSE score	29.47 ± 0.74	28.44 ± 1.61***	29.00 ± 1.26	28.47 ± 1.59*	29.17 ± 1.33	28.10 ± 1.73***
Postinjury days (initial scan; d)	N.A.	6.76 ± 2.85	6.18 ± 2.96	5.63 ± 3.32	5.00 ± 1.73	5.10 ± 3.14
Postinjury days (follow-up; d)	N.A.	N.A.	374.82 ± 15.75	373.26 ± 14.31	751.6 ± 26.25	735.50 ± 9.65
Clinical Characteristics						
PSQI	4.86 ± 2.20	8.07 ± 4.36***	5.27 ± 3.44 †	7.47 ± 3.42**	5.67 ± 4.55	8.60 ± 3.66***
PCSQ score	4.94 ± 5.22	13.30 ± 13.43***	0.91 ± 0.94 ††	13.74 ± 8.93***†††	1.00 ± 0.89 †	15.20 ± 12.12***††
BAI score	3.91 ± 4.32	9.80 ± 9.57***	4.09 ± 4.18	9.11 ± 8.92**	1.50 ± 3.00	10.89 ± 8.02**†
BDI score	4.56 ± 4.61	9.70 ± 9.86**	3.73 ± 4.96	8.84 ± 6.14**†	3.50 ± 5.74	10.87 ± 5.56**††

Note.—Demographic or behavioral characteristics compared between patients with mTBI and HCs (* $P < .05$, ** $P < .01$, *** $P < .001$), between the initial and follow-up data of patients with mTBI († $P < .05$, †† $P < .01$) and between symptom-resolved and symptom-prolonged groups (‡ $P < .05$, ‡‡ $P < .01$, ‡‡‡ $P < .001$). Data presented as means ± standard deviations.

BAI = Beck Anxiety Inventory; BDI = Beck Depression Inventory; GOSE = Glasgow Outcome Scale-Extended; HC = healthy control; MMSE = Mini-Mental State Examination; mTBI = mild traumatic brain injury; PCSQ = Rivermead Post Concussion Symptoms Questionnaire; PSQI = Pittsburgh sleep quality index; RT = reaction time.

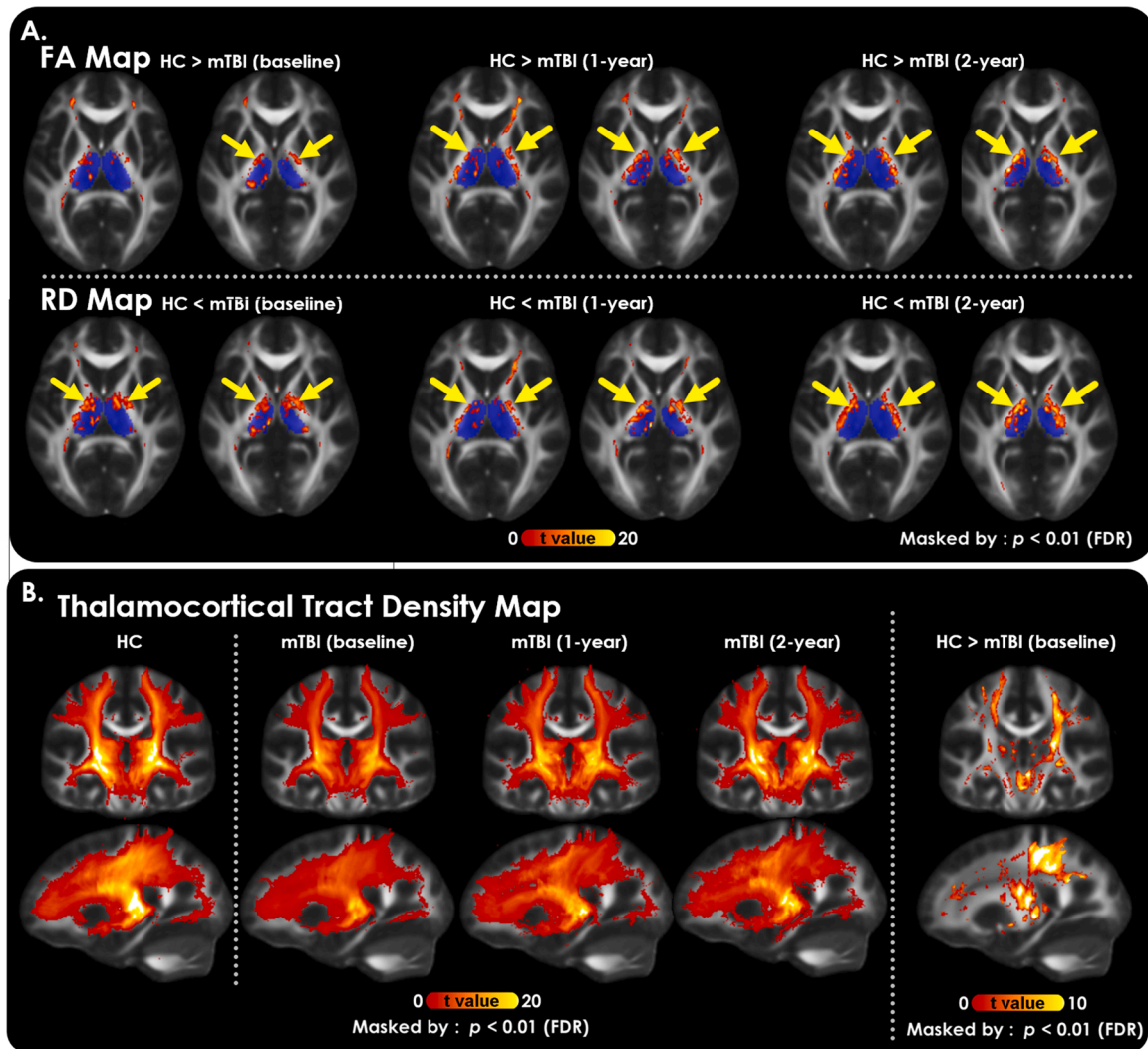


Fig. 2. Concussion-induced shear forces underlie thalamic reticular nucleus (TRN) and thalamocortical tract injury development after concussion. (A) Significantly decreased fractional anisotropy (FA; top) and increased radial diffusivity (RD; bottom) at the bilateral thalamic borders (yellow arrows) were observed in patients with mTBI compared with healthy controls (HCs; false discovery rate [FDR]-corrected $P < .01$) at baseline as well as 1- and 2-year follow-ups. Translucent dark blue ovals indicate locations of the bilateral thalami. These maps were further masked by the threshold of the group-averaged FA (>0.2) to ensure that only the white matter tracts remained. (B) A significant reduction in thalamocortical tract density was observed in the mTBI group compared with the HC group (FDR-corrected $P < .01$) only at baseline. The top row presents the coronal slice passing through the center of the bilateral thalami; the bottom row indicates the sagittal slice passing through the center of the left thalamus.

Table 3

Cluster Statistics for Significant Differences in DTI-Parametric Maps near the Peri-Thalamic Thalamic Reticular Nucleus (TRN) Region between HCs and Patients.

	DTI parameter	Side	Area (mm ³)	T-statistics	MNI coordinate		
					X	Y	Z
baseline	RD	L	4396	7.998	-11	-10	16
		R	2585	6.748	12	-8	5
1-year	FA	L	1623	7.261	-10	-11	17
		R	203	5.396	16	-18	18
	RD	L	3864	9.456	-20	-28	16
		R	2112	6.334	21	-16	0
2-year	FA	L	3715	8.064	-25	-25	8
		R	2078	7.067	25	-15	11
	RD	L	4736	7.998	-11	-10	16
		R	1616	6.748	12	-8	5
	FA	L	5319	10.385	-21	-19	1
		R	3903	9.772	25	-14	12

particularly in the double CHI rats, as observed through visual

comparison. These findings confirm the varying extents of myelin damage following injury. Similarly, the decreased MBP immunoreactivity at the TRN could still be observed at 7 weeks after CHI, whereas damage near the cortical fiber disappeared, suggesting the long-lasting axonal damage, myelin damage, or both in the TRN after CHI.

3.3. Disrupted thalamocortical coherence after concussion

3.3.1. Clinical human results

Patients exhibited resting state hyperconnectivity among their thalamic subdivisions (Fig. 3A). In other words, the fMRI signal in different thalamic subdivisions became similar after concussion (Fig. 3G), which may imply the failure of thalamic function in relaying and integrating the different information. This significant elevation in within-thalamic functional connectivity was also observed at the 1- and 2-year follow-ups in patients after concussion (Fig. 3A)—indicating no significant restoration of the disrupted thalamic function in patients.

Furthermore, we observed an increase in rhythmic coherence between the thalami and almost all cortical regions (Fig. 3B) in the

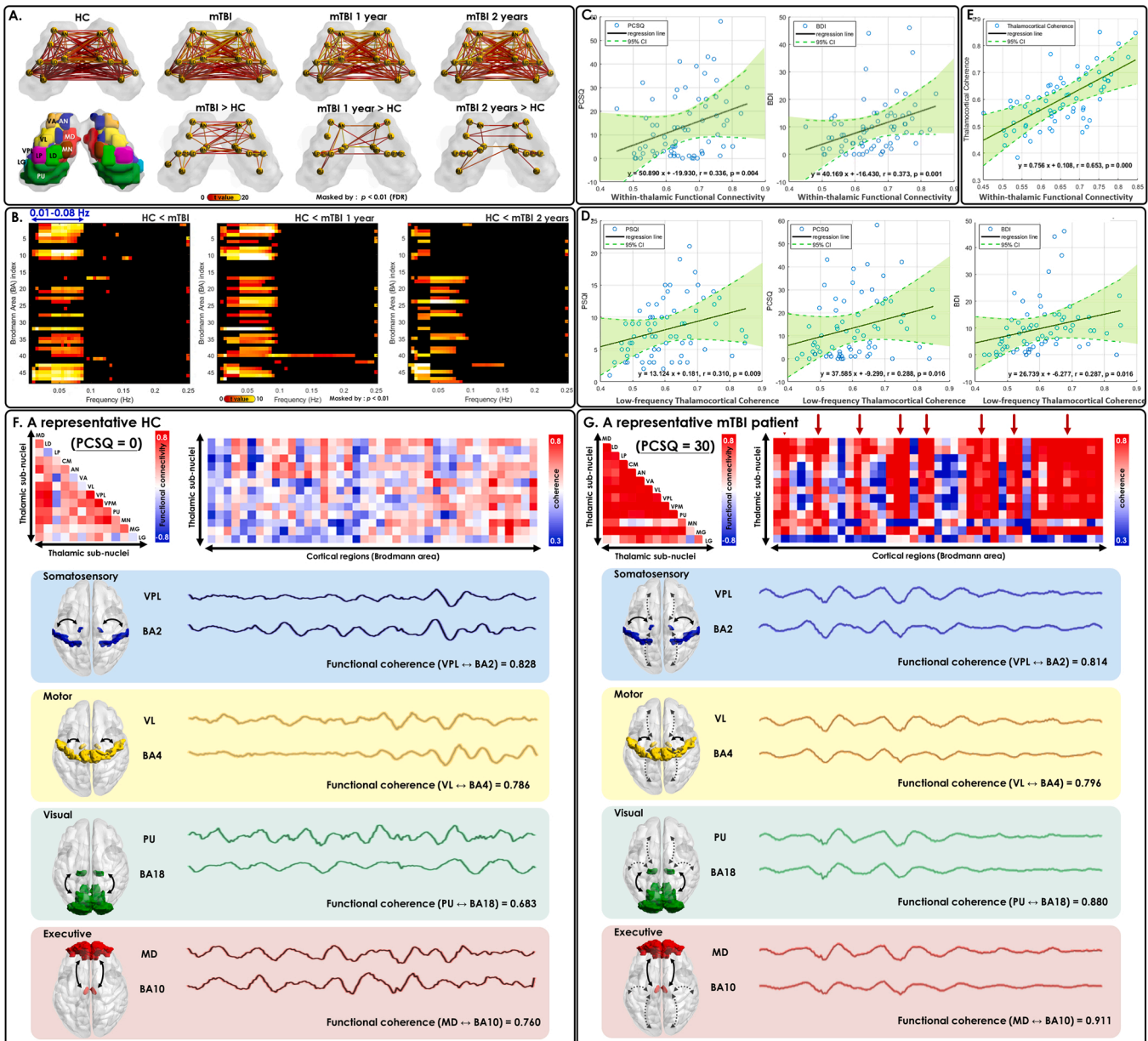


Fig. 3. Functional connectivity and temporal coherence changes are associated with clinical symptoms after concussion. (A) Patients exhibiting significantly increased within-thalamic resting-state functional connectivity compared with the healthy controls (HCs; false discovery rate [FDR]-corrected $< .01$) in baseline assessment and follow-ups. (B) Increase in coherence between thalami and almost all cortical regions in the low-frequency band (disease-dominant frequency range = 0.01–0.08 Hz) in patients compared with HCs ($P < .01$) in baseline assessment and follow-ups. (C) Averaged within-thalamic functional connectivity positively correlated with Rivermead Post Concussion Symptoms Questionnaire (PCSQ) scores ($r = 0.336, P = .005$) and Beck Depression Inventory (BDI) scores ($r = 0.372, P = .002$) in patients at the initial visit. (D) Averaged low-frequency thalamocortical coherence was positively correlated with PCSQ scores ($r = 0.288, P = .016$), and BDI scores ($r = 0.287, P = .016$; right) in patients assessed at the initial visit. (E) Significantly positive correlation between averaged within-thalamic hyperconnectivity and averaged low-frequency thalamocortical coherence ($r = 0.653, P < .001$) was noted in the patient group in the baseline assessment. Solid black lines in (C), (D), and (E) represent the estimated linear regression lines; translucent light green areas surrounded by green dashed lines indicate 95% confidence interval of the estimated linear regression line. (F) Within-thalamic functional connectivity matrix (left) and thalamocortical functional coherence matrix (right) for a representative HC. (G) Within-thalamic functional connectivity matrix (left) and thalamocortical functional coherence matrix (right) for a representative patient with mild traumatic brain injury (mTBI). Red arrows denote the abnormal patterns of pathologically elevated low-frequency thalamocortical rhythmicity, showing a high degree of temporal coherence with almost all cortical regions. fMRI signals were band-pass filtered to 0.01–0.08 Hz before functional connectivity and coherence calculations in (F) and (G).

frequency range = 0.01–0.08 Hz (disease-dominant frequency range; $p < .01$ [FDR]). Specifically, the globally enhanced thalamocortical coherence interfered with the cortical regions in the different thalamocortical loops (Fig. 3G; red arrows) indicated the fMRI signal between different thalamocortical loops became similar after concussion (Fig. 3G), thus disrupted the thalamocortical multifunctionality. This was also evident in our patients at the 1- and 2-year follow-ups after

concussion (Fig. 3B)—suggesting no significant restoration of the disrupted thalamocortical function in patients.

As mentioned above, because of the highly idiosyncratic nature of concussions, the concussion-induced thalamic and/or thalamocortical damage can vary from person to person. The within-thalamic functional connectome matrix (Fig. 3F,G; left) was then simply averaged as a compromise strategy to individually quantify the degree of elevation in

within-thalamic functional connectivity and yield the potential personalized disease burden of thalamic dysfunction. Similarly, the thalamocortical coherence matrix (Fig. 3F,G: right) was also averaged as a compromise strategy to individually quantify the degree of pathologically increased thalamocortical coherence. Notably, the averaged within-thalamic connectivity at baseline was significantly elevated in the patient group (HC: 0.23 ± 0.15 , mTBI: 0.43 ± 0.12 , $p < .0001$ [FDR]) and correlated with patient PCSQ scores (Fig. 3C: left) and BDI scores (Fig. 3C: right). The averaged thalamocortical coherence across all cortical regions was significantly enhanced in the patient group (HC: 0.47 ± 0.07 , mTBI: 0.60 ± 0.10 , $p < .0001$ [FDR]) and correlated with clinical symptoms, including sleep quality (PSQI; Fig. 3D: left), PCS (PCSQ score; Fig. 3D: middle), and depression (BDI score; Fig. 3D: right)—suggesting the close relationship between these two connectome biomarkers and PCS.

3.3.2. Translational rodent results

Similar to the clinical results of our patients (Fig. 3), CHI rodents also exhibited resting-state hyperconnectivity among their thalamic subdivisions (Fig. 4A,B). This thalamic dysfunction was observed as early as the 1st week after CHI and persisted for up to 7 weeks (Fig. 4A,B),

indicating prolonged disruption of thalamic function in CHI rodents. The increase in rhythmic coherence between the thalami and almost all cortical regions was also evident in the CHI rodents (Fig. 4C,D). Importantly, this pathological coherent rhythmic change resolved 3 weeks after single CHI, but after double CHI, it persisted for up to 7 weeks (Fig. 4D), confirming the close relationship of disrupted thalamocortical coherence with disease severity and prolongation.

3.4. The relationship between TRN Injury and disrupted thalamocortical coherence

The human TRN is a very thin layer (in submillimeter scale (Gringel et al., 2009)) wrapped the dorsolateral and anterior segments of the thalamus, thus its injury is difficult to be individually quantified via clinical DTI (slice thickness ~ 3 mm) in 3 T MR scanner. Fortunately, this microstructural injury can be detected and quantified at rodents in 7 T small animal MR scanner. The degree of TRN injury in the CHI rodents was significantly correlated with the enhancement in within-thalamic functional connectivity assessed at 1 week after CHI (Fig. 4E), providing translational evidence that the concussion-induced thalamocortical dysfunction is linked to the different levels of peri-thalamic TRN

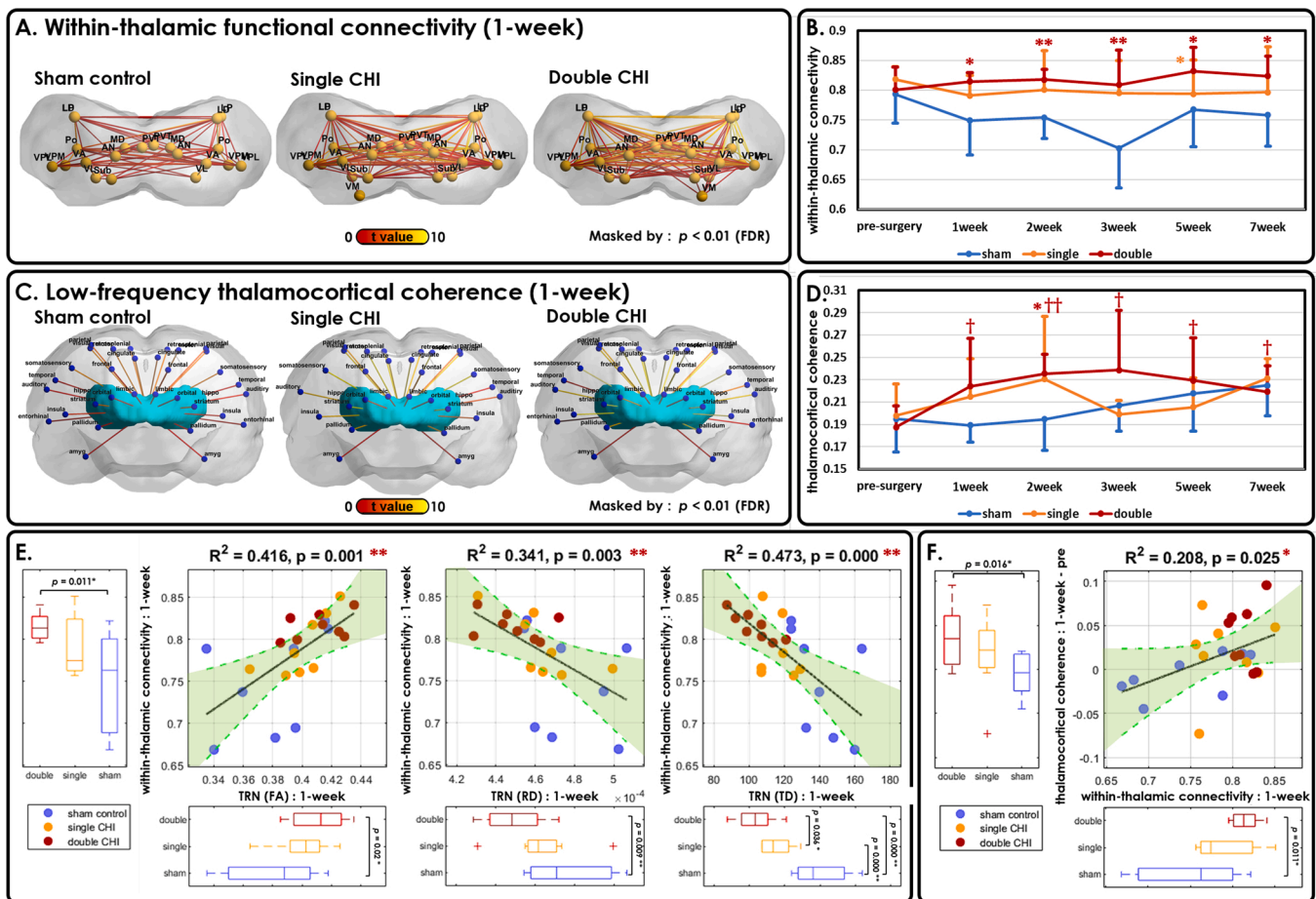


Fig. 4. Within-thalamic functional connectivity and low-frequency thalamocortical temporal coherence change after sham, single, and double closed-head injury (CHI) surgery. The single (middle) and double (right) CHI groups exhibited significantly increased (A) within-thalamic functional connectivity and (C) low-frequency (0.01–0.08 Hz) thalamocortical coherence compared with the sham group (left) 1 week after surgery (false discovery rate [FDR]-corrected $P < .01$). Longitudinal changes in (B) averaged within-thalamic functional connectivity and (D) averaged low-frequency thalamocortical coherence during the first 7 weeks after surgery (* $P < .05$, ** $P < .01$ compared with sham group; compared with pre-surgery: † $P < .05$, †† $P < .01$). (E) Averaged within-thalamic functional connectivity was significantly correlated with fractional anisotropy (FA; $R^2 = 0.416$, $P = .001$; left), radial diffusivity (RD; $R^2 = 0.341$, $P = .003$; right), and track density (TD; $R^2 = 0.473$, $P < .001$; right) within the thalamic reticular nucleus (TRN) 1 week after injury. (F) Significant positive correlation was noted between averaged within-thalamic functional connectivity and averaged low-frequency thalamocortical coherence ($R^2 = 0.208$, $P < .025$). Solid black lines in (E) and (F) represent the estimated linear regression lines; translucent light green areas surrounded by green dashed lines indicate the 95% confidence interval of the estimated linear regression line.

injury.

3.5. Roles of TRN Injury and disrupted thalamocortical coherence in PCS prolongation

3.5.1. Clinical human results

Table 1 presents the PCS progression estimated via PSQI, PCSQ scores, and BDI scores (Fig E6A–C [online]) of our patients. Table 2

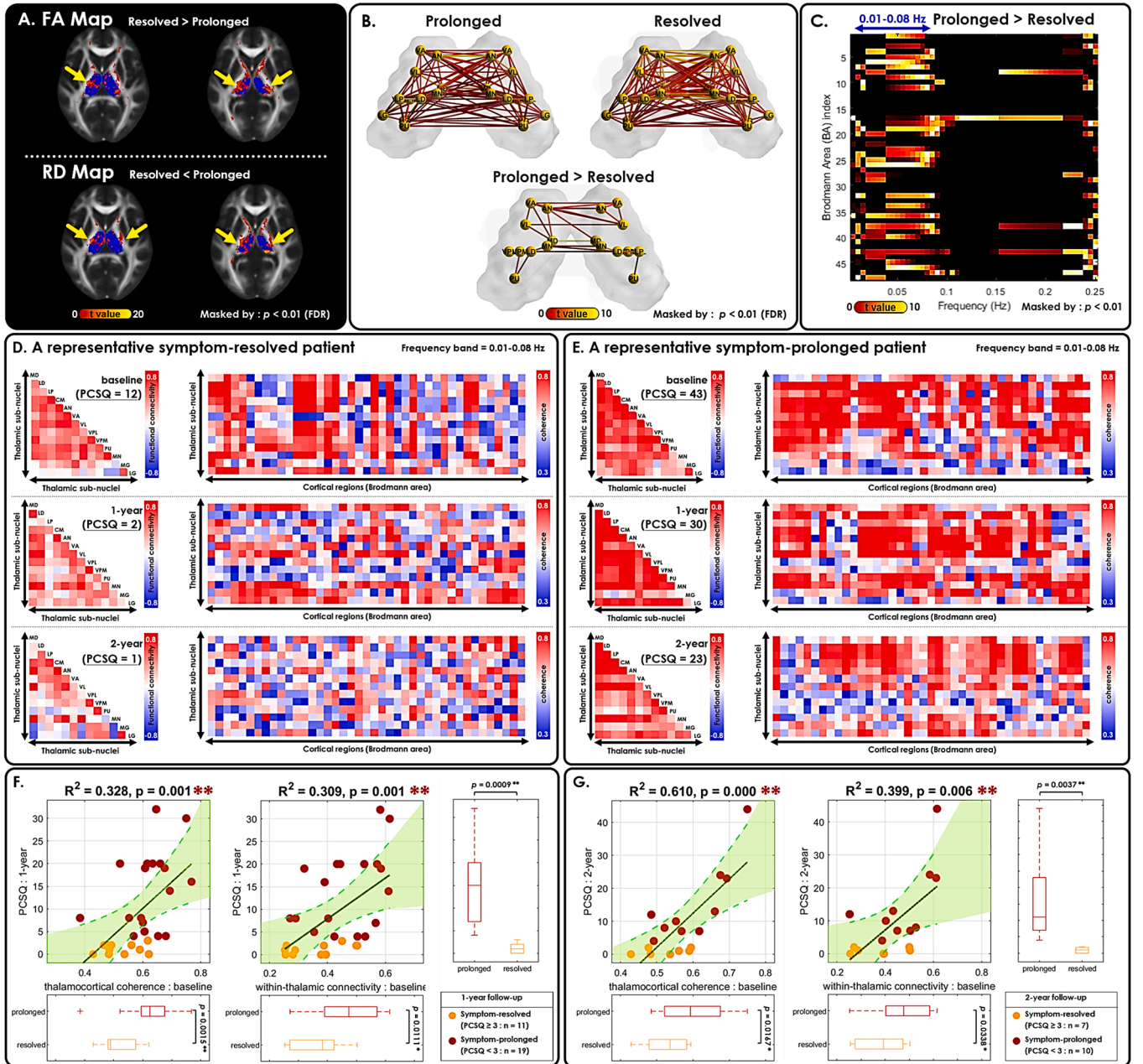


Fig. 5. Degrees of TRN injury the pathological low-frequency thalamocortical rhythmic coherence at baseline differ between patients with resolved or prolonged postconcussive symptoms (PCS) at 1-year follow-up. (A) Significantly lower fractional anisotropy (FA; top row) and higher radial diffusivity (RD; bottom row) near the peri-thalamic thalamic reticular nucleus (TRN; false discovery rate [FDR]-corrected $P < .01$) in the symptom-prolonged group compared with the symptom-resolved group at baseline. (B) Higher within-thalamic functional connectivity (FDR-corrected $P < .01$) in the symptom-prolonged group (right) compared with the symptom-resolved group (left) at baseline. (C) Increase in coherence between the thalami and almost all cortical regions in the low-frequency band (0.01–0.08 Hz) of the symptom-prolonged group compared with that of the symptom-resolved group ($P < .01$) at baseline. (D) Within-thalamic functional connectivity matrix (left) and thalamocortical functional coherence matrix (right) for a representative symptom-resolved patient at baseline (top), 1-year (middle), and 2-year (bottom) assessment. (E) Within-thalamic functional connectivity matrix (left) and thalamocortical functional coherence matrix (right) for a representative symptom-prolonged patient at baseline (top), 1-year (middle), and 2-year (bottom) assessment. fMRI signals were band-pass filtered to 0.01–0.08 Hz before calculating functional connectivity and coherence in (D) and (E). The long-term PCS after (F) 1 year and (G) 2 years are correlated with thalamocortical dysrhythmia (TCD)-related connectome biomarkers, including the averaged low-frequency thalamocortical coherence, and averaged within-thalamic hyperconnectivity. Red dots indicate symptom-prolonged patients (Rivermead Post Concussion Symptoms Questionnaire [PCSQ] score ≥ 3); orange dots represent symptom-resolved patients (PCSQ score < 3).

further divides the symptom-resolved and symptom-prolonged patients according to a published cut-off score (PCSQ ≥ 3) (Sigurdardottir et al., 2009). The symptom-prolonged patients demonstrated significantly lower FA but higher RD in peri-thalamic TRN regions (Fig. 5A), higher within-thalamic functional connectivity (Fig. 5B), and enhanced thalamocortical coherence (Fig. 5C) at baseline compared with the symptom-resolved patients. This result suggested that acute TRN injury and disrupted thalamocortical coherence may be associated with PCS prolongation.

At both 1- and 2-year follow-ups, the symptom-resolved and symptom-prolonged patients exhibited distinct patterns of baseline biomarkers (i.e., the within-thalamic functional connectivity [1-year (resolved): 0.36 ± 0.09 , 1-year (prolonged): 0.47 ± 0.11 , $p = .011$ (FDR); 2-year (resolved): 0.36 ± 0.09 , 2-year (prolonged): 0.50 ± 0.09 , $p = .034$ (FDR)] and thalamocortical coherence [1-year (resolved): 0.52 ± 0.06 , 1-year (prolonged): 0.62 ± 0.09 , $p = .001$ (FDR); 2-year (resolved): 0.53 ± 0.06 , 2-year (prolonged): 0.62 ± 0.09 , $p = .016$ (FDR)]); both of which were strongly associated with the degree of long-term symptoms (Fig. 5D-G). The GLM analyses showed a significant relationship between the severity of long-term PCS and the two baseline predictor variables (i.e., within-thalamic functional connectivity [1-year: $\beta = 31.94$, $p = .046$; 2-year: $\beta = 30.57$, $p = .019$] and thalamocortical coherence [1-year: $\beta = 20.58$, $p = .037$; 2-year: $\beta = 81.95$, $p = .023$]), by controlling for confounding factors (i.e., age [1-year: $\beta = 0.16$, $p = .34$; 2-year: $\beta = 0.17$, $p = .49$] and sex [1-year: $\beta = 2.45$, $p = .49$; 2-year: $\beta = 3.31$, $p = .51$]). The models explained 45.3% of the variance in PCSQ assessed at the 1-year follow-up ($R^2 = 0.453$) and 66.7% of the variance in PCSQ obtained at the 2-year follow-up ($R^2 = 0.667$), respectively. These pathological patterns in coherence matrix not only aided in dichotomizing patients with prolonged and resolved symptoms (Fig. 5DE) but also serve to correlate with the trajectory of the symptom progression in patients with prolonged symptoms and resolved symptoms (Fig. 5DE: middle and bottom), suggesting that as thalamocortical coherence is restored, so should one's PCS.

3.5.2. Translational rodent results

In rodents, long-lasting TRN injury was confirmed through DTI and immunohistochemistry (Fig E3 and E4 [online]). The long-term behavioral deficits assessed at 7 weeks after single CHI were significantly correlated with baseline neuroimaging biomarkers (Fig E5 [online]), including the extent of TRN injury and increased within-thalamic hyperconnectivity, and thalamocortical coherence. In the double CHI group, worse behavioral outcomes such as elevated mNSS (Fig E5 [online]), prolonged beam walking duration (Fig E5B [online]), and worse recognition memory performance (Fig E5C [online]) were observed from 24 h to 7 weeks after concussion measurements. Considering that the two CHIs were administered only 1 h apart in the double CHI group, these results suggest cumulative injury after repeated impacts within a short period (Kao et al., 2019), and the injury-induced behavioral deficits can persist for ≥ 7 weeks. Our results from rats provide translational evidence for the close relationship between baseline neuroimaging biomarkers and prolonged PCS in humans.

In addition to TRN injury, thalamocortical tract injuries may play a role in PCS prolongation. However, our results demonstrated that the thalamocortical tract injury revealed through DTI were only mild at the 1-year follow-up in patients (Figs. 2) and 3 weeks after CHI in rats (Fig E3 [online]). Furthermore, Fig E7 [online] demonstrated the nonsignificant correlation of the long-term behavioral deficits assessed 7 weeks after CHI with the severity of thalamocortical tract damage between the thalamus and impacted left primary somatosensory cortex of the hindlimb region (S1HL) assessed within 1 week of CHI; this suggested that the thalamocortical tract damage is not the main contributor of PCS prolongation.

3.6. Predictive power and external validity of fMRI-based coherence biomarkers for long-term PCS prediction

To test the generalizability of the fMRI-based biomarkers proposed in this study, the baseline resting-state fMRI and longitudinal PCSQ data of 10 patients with mTBI were collected from another external patient cohort from Shuang-Ho Hospital (SHH; Table 4), and the predictive power of the baseline biomarkers for PCSQ at the 1- and 2-year follow-ups was analyzed. The results demonstrated strong correlation between the connectome biomarkers and PCS at 1- and 2-year follow-ups (Fig. 6A-D), even though MRI acquisition parameters were not entirely identical between the internal (TMUH) and external cohorts (Table 5). The GLM analyses showed a significant relationship between the severity of long-term PCS and the two baseline predictor variables (i.e., within-thalamic functional connectivity [1-year: $\beta = 84.21$, $p = .049$; 2-year: $\beta = 10.046$, $p = .036$] and thalamocortical coherence [1-year: $\beta = 101.96$, $p = .009$; 2-year: $\beta = 43.83$, $p = .002$]), by controlling for confounding factors (i.e., age [1-year: $\beta = -0.02$, $p = .708$; 2-year: $\beta = -0.02$, $p = .276$] and sex [1-year: $\beta = 0.46$, $p = .80$; 2-year: $\beta = 0.55$, $p = .34$]). The models explained 90.4% of the variance in PCSQ assessed at the 1-year follow-up ($R^2 = 0.904$) and 97.1% of the variance in PCSQ obtained at the 2-year follow-up ($R^2 = 0.971$), respectively. These results thus confirmed the external validity and stability of these two connectome biomarkers.

Both the baseline biomarkers, along with the patient age and sex, were next used to train a machine learning-based long-term PCS predictive model using TMUH data and testing by the external SHH cohort data. It yielded high predictive powers for persistent PCS in not only internal but external cohort data at the 1- and 2-year follow-ups (Fig. 6E, F), again confirming the predictive power of the baseline biomarkers for long-term PCS (1- and 2-year PCSQ score).

In addition, this machine learning-based predictive strategy was also applied to the rodent data, which could also satisfactorily predict the long-term behavioral deficits (ie, neurological function, balance function, and recognition memory; Fig E8 [online]), further providing the translational evidence that these biomarkers can facilitate early prediction of long-term PCS.

4. Discussion

In this study, we demonstrated the interplay between persistent peri-thalamic TRN injury and disrupted thalamocortical coherence and

Table 4
Demographic Characteristics and Longitudinal Postconcussive Symptoms in Individuals in Internal (TMUH) and External (SHH) Patient Cohorts.

Demographic Characteristics	TMUH Cohort (1-y Follow-Up)	TMUH Cohort (2-y Follow-Up)	SHH Cohort (1- and 2-y Follow-Ups)
Sample size (n)	30	17	10
Sex (female/male)	22/8	9/8	7/3
Age (y)	36.80 ± 11.29	35.76 ± 10.97	36.10 ± 15.55
Education (y)	14.65 ± 2.76	14.71 ± 2.82	14.70 ± 1.83
Postinjury days (initial scan; d)	5.83 ± 3.15	5.29 ± 2.69	$11.80 \pm 7.67^*$
Clinical Characteristics			
PCSQ score	9.03 ± 9.45	9.29	2.40
PCS (prolonged/resolved)	19/11	± 11.66 10/7	$\pm 2.91^{**}$ 7/3
			$\pm 2.91^*$ 7/3

Note.—Demographic characteristics or symptoms compared between the TMUH and SHH cohorts (* $P < .05$, ** $P < .01$). Data presented as means $\pm$ standard deviations. HC = healthy control; mTBI = mild traumatic brain injury; PCSQ = Rivermead Post Concussion Symptoms Questionnaire; SHH = Shuang-Ho Hospital; TMUH = Taipei Medical University Hospital

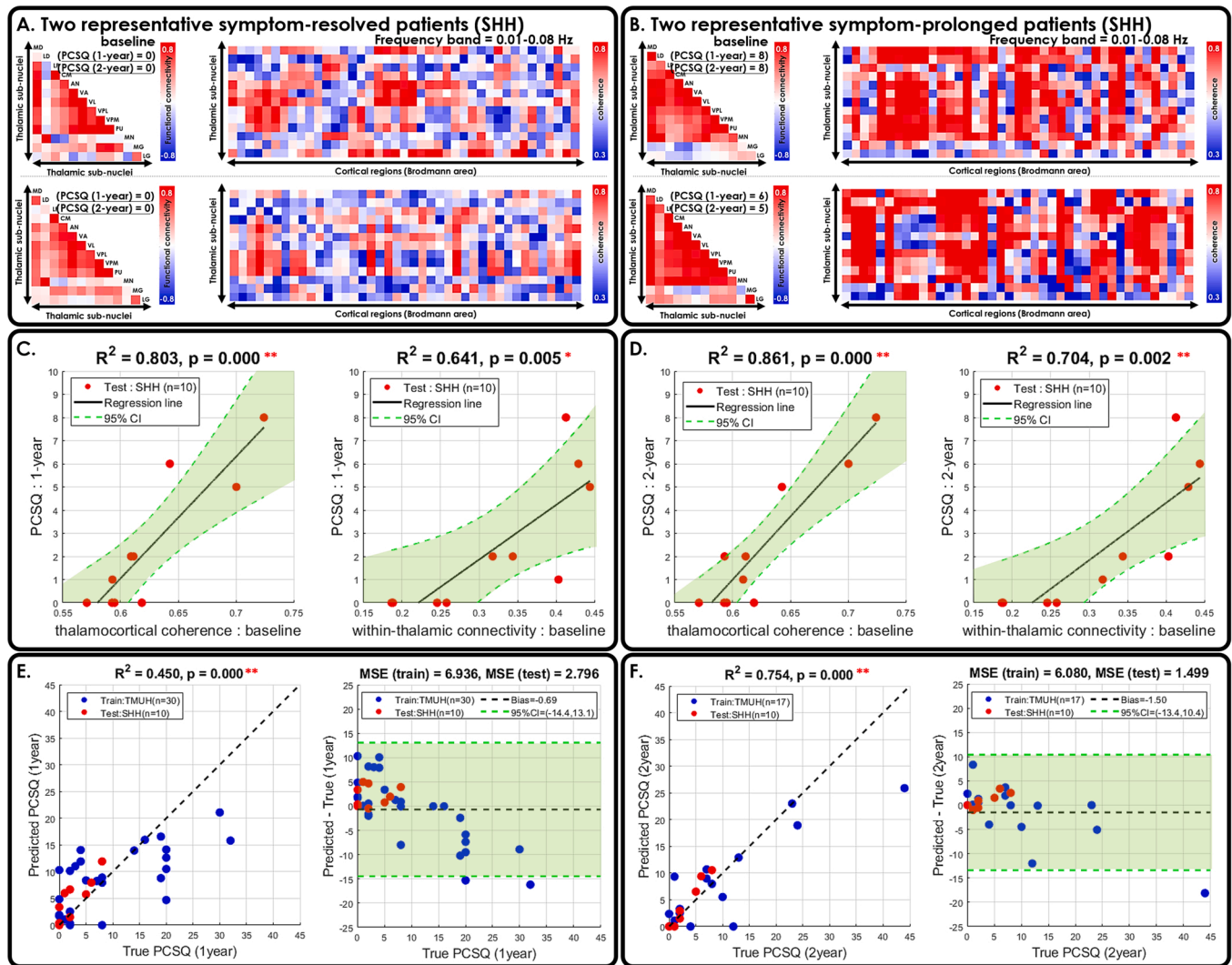


Fig. 6. External validation of the baseline fMRI-based connectome biomarkers. Connectome biomarkers, including within-thalamic functional connectivity matrix (left) and thalamocortical functional coherence matrix (right) for two representative (A) symptom-resolved patients and two representative (B) symptom-prolonged patients from Shuang-Ho Hospital (SHH) collected at baseline. The long-term postconcussive symptoms (PCS) in patients from SHH after (C) 1 year and (D) 2 years are significantly correlated with the baseline connectome biomarkers, including the averaged low-frequency thalamocortical coherence and averaged within-thalamic functional connectivity. Solid black lines in (C) and (D) represent the estimated linear regression lines; translucent light green areas surrounded by green dashed lines in (C) and (D) indicate the 95% confidence interval of the estimated linear regression line. Long-term PCS at 1-year (E) and 2-year (F) follow-ups satisfactorily predicted by the machine learning-based predictive model trained by the two connectome biomarkers and patient age and sex in the Taipei Medical University Hospital (TMUH; ie, internal) cohort and then tested using SHH (ie, external) cohort data. The scatter plots (left) and Bland-Altman plots (right) of the internal (TMUH; blue dots) and external (SHH; red dots) predictions are presented in (E) and (F). Dashed black lines in the right panel of (E) and (F) represent the prediction bias in training data; translucent light green areas surrounded by green dashed lines in the right panel of (E) and (F) indicate the 95% confidence interval of the prediction bias in training data.

Table 5
Comparison of Human MRI Acquisition Parameters for TMUH and SHH Cohorts.

fMRI Acquisition Parameters	TMUH	SHH
MRI branch	Siemens MAGNETOM Prisma	GE Discovery MR 750
Field strength (T)	3.0	3.0
Receive coil	64-channel head coil	8-channel head coil
TR/TE (ms)	2000/20	2000/35
Flip angle (°)	90	90
Voxel size (mm ³)	3 × 3 × 3.5	3.5 × 3.5 × 4
Matrix size	64 × 64 × 40	64 × 64 × 40
Resting state (volumes)	190	270
Resting state (total acquisition time)	6'20"	9'00"

Note.—fMRI = functional MRI; SHH = Shuang-Ho Hospital; TE = echo time; TMUH = Taipei Medical University Hospital; TR = repetition time

clarified their relationship with prolonged PCS. Concussion can predispose patients to varying degrees of peri-thalamic TRN injury, leading to different increment levels of within-thalamic and thalamocortical functional coherence, manifesting PCS of varied severity. These thalamocortical functional disruptions can be monitored at the individual level by using biomarkers derived from the fMRI-based coherence matrix, which can predict the long-term PCS in both human and rodent. Taken together, the current results may aid in delineating the chain of events beginning with TRN injury and resulting in a surge in disrupted thalamocortical coherence, leading to PCS prolongation. Our systematic analyses contributed to the understanding of the involvement of impaired thalamocortical coherence system after concussion. It also presented a framework for precise personalized prediction of prolonged PCS, which performed well in both internal and external patient cohorts. Disrupted thalamocortical coherence observed by fMRI-based

biomarkers may provide a new avenue for prolonged PCS diagnosis, prognosis, and treatment. Our future studies are underway to investigate the relationship between the varying connectome matrix signatures and the various type of PCS.

4.1. Possible pathogenetic mechanisms underlying peri-thalamic TRN injury and disrupted coherence after concussion

TRN, the only GABAergic thalamic subdivision, may serve as the brain pacemaker, which modulates thalamocortical synchrony (Min, 2010). In particular, neuronal dysfunction of the TRN can result in disinhibition of thalamic relay cells (Marini et al., 2002); this can lead to the abnormal synchronization of irrelevant cortical network and thereby interfere with the information transmission between appropriately synchronized cortical neurons, eventually causing thalamocortical dysrhythmia (TCD) (Llinás et al., 1999). Conventionally, electroencephalography (EEG) and magnetoencephalography (MEG) are most frequently used to investigate the disturbed thalamocortical oscillation mechanisms in TCD (Llinás et al., 1999; Llinás and Steriade, 2006). The abnormally increased within-thalamic functional connectivity and thalamocortical coherence can nevertheless also be investigated through fMRI—as indicated by our results. This enhancement in within-thalamic functional connectivity and thalamocortical coherence has also been previously noted in patients with schizophrenia (Baran et al., 2019), Parkinson disease (Owens-Walton et al., 2019) and acute mTBI (Sours et al., 2015; Zhou et al., 2014); it can be attributed to the pathological rhythmic coupling from thalamic loops with different functions (Llinás et al., 1999). Our findings can also be compared with previous studies stating that the wide spread low-frequency (theta wave observed by EEG and/or MEG) thalamic rhythmicity can disrupt thalamocortical functional interactions, thereby producing negative symptoms noted in TCD such as sleep disorder, attentional deficits, fatigue, and depression (Llinás et al., 1999; Llinás and Steriade, 2006). Further, our observed positive correlation between elevated within-thalamic functional connectivity and thalamocortical coherence seem to provide additional evidence for Llinás and colleagues' (2005) notion of a predominant, pathological low-frequency thalamic rhythmicity in TCD patients coupling with the thalamocortical oscillatory system, which we think is responsible for the fMRI results in the present study. This predominant slow rhythmicity then leads to, and shows positive correlations with, symptoms such as sleep quality, PCS, and depression (Llinás et al., 1999; Llinás and Steriade, 2006)—consistent with our results. However, the relationship between the role of the fMRI-based biomarkers noted in our study and TCD biomarkers proposed by previous EEG and/or MEG studies remains unclear because our fMRI sampling rate may be too broad ($TR = 2$ s) and the neurovascular-coupled responses may wrap the neuronal oscillations into the low-frequency band (Lin et al., 2018). Thus, we emphasize that the pathologically enhanced thalamocortical coherence delineated through fMRI in this study may not be directly translated to the electrophysiological phenomena and neurobiological mechanisms. Further investigation using simultaneous EEG-fMRI to link EEG biomarkers with fMRI biomarkers in mTBI is needed to help unravel the complete mechanism underlying TCD and more precisely predict disease-related symptom progression.

4.2. Functional connectivity and temporal coherence of thalamocortical rhythmicity in mTBI

The difference between functional connectivity and functional coherence can be observed in terms of signal transmission latency and hemodynamic response shape (Sun et al., 2004). In particular, functional connectivity estimated through simple correlational analyses with zero-time lag is sensitive to interregional signal transmission latencies and hemodynamic response shapes (Li et al., 2021; Lin et al., 2018; Sun et al., 2004), but temporal coherence is not. As such, if two brain regions demonstrate identical underlying neural activities—but with distinct

hemodynamic responses, different signal transmission latencies, or both—their time series can reflect correlation differences but with high coherence (Sun et al., 2004). Therefore, on the basis of our findings, we suggest that thalamocortical network disruption in mTBI results from a combined effect of 1) inefficient signal transmission due to axonal damage, myelin loss, or both in the thalamocortical tracts and 2) pathologically enhanced thalamocortical coherence. This may also be the reason for the somewhat conflicting results noted in previous studies: a study reported a decrease in the thalamocortical functional connectivity of patients with mTBI (Banks et al., 2016), whereas two other studies reported an increase in temporal coherence or latency corrected-functional connectivity in patients with mTBI (Sours et al., 2015; Zhou et al., 2014). The specific reasons for the differences in these findings may be related to the analytical approaches used in the studies, which may have probed different mechanisms of the disturbed thalamocortical coherence system after concussion.

4.3. Limitations

4.3.1. Dropouts in longitudinal data

The early biomarkers proposed in our study and the PCS did not improve significantly at the 1- and 2-year follow-ups, confirming that even mTBI can cause widespread, potentially long-lasting effects (Moeller et al., 2011). However, the considerable dropouts in our longitudinal data should also be considered, so the results can also be explained by attrition bias, meaning that perhaps only patients who still felt uncomfortable after 1 or 2 years continued participating in this study. Although the outcomes of every lost patient or why they left the trial are impossible to know, we could at least confirm that a subset of patients may experience persistent PCS that may be associated with peri-thalamic TRN injury and disrupted thalamocortical coherence.

4.3.2. Control the consciousness state during resting-state fMRI scan

Resting state fMRI is highly susceptible to participant-induced variability. In human, we used several pre-scan instructions to help ensure that the participants were in a relaxed, stable, and alert state during the scan. Specifically, participants were advised to get adequate rest and avoid intense physical activity before the scan to avoid fatigue or drowsiness that might interfere with the resting-state measurement (Lee et al., 2013; Smitha et al., 2017). However, we did not monitor or control the consciousness state of the participants during the scan. It is possible that increased drowsiness in patients with PCS may have contributed to our findings. For the rodents, we used a previously published protocol to prepare the animals (Hsu et al., 2016). However, it is important to note that the protocol only minimizes, but cannot entirely eliminate the interference of anesthesia. Prior to scanning, the animals received a 0.03 mg/kg, subcutaneous, bolus injection of dexmedetomidine, followed by continuous infusion at 0.015 mg/kg/h, while the level of isoflurane was gradually reduced to 0.5% (Hsu et al., 2016). Resting-state fMRI data were collected approximately 90 min after the initiation of dexmedetomidine infusion, at which time the resting-state fMRI signal was maximally stable (Hsu et al., 2016; Lu et al., 2012). Future studies need to carefully control the state of consciousness in both participants and rodents to enhance the accuracy and stability of resting-state fMRI measurements.

4.3.3. The cross-species differences and generalizability of animal model to mimic clinical status after concussion

Here, we used a translational rodent CHI model, which could produce peri-thalamic TRN injury from impact-acceleration shearing forces similar to that in patients. However, even though the general trends regarding postconcussive alterations were similar, the process of the concussive injury and the results might not entirely identical to that noted in human pathology. Specifically, we used a fixed target impact region in our weight-drop model to control the experimental conditions, and two repetitive impacts apart with a 1-hour interval to modulate the

impact severity and avoid focal hemorrhage or contusion. The experimental conditions included here may be considerably different from clinical scenarios in which variations in the impacted regions often occur and could be essential for specific symptomatic and behavioral changes. In addition, only male rats were included in our rodent study. Previous studies have suggested that the male rats were used as the standard because the researchers believed that hormonal changes throughout the female cycle caused female rats to react differently to the same stimuli (Beery and Zucker, 2011; Wildner, 2019). This made their responses unpredictable and led to variable results during repeated experiments (Beery and Zucker, 2011; Wildner, 2019). Therefore, we used only male rats to reduce the variability caused by hormone fluctuation and increase the reproducibility of the findings. However, we acknowledge that excluding female rats may limit our understanding of how certain pathogenetic mechanisms affect both sexes. Further studies are needed to replicate these experiments using female rats to investigate the potential sex differences in this pathogenetic mechanism. Nevertheless, our rodent model could recapitulate our human study in that the persistent damage to peri-thalamic TRN, even in a very mild degree of injury, triggering the thalamocortical coherence disruptions, ultimately inducing prolonged PCS.

4.3.4. Age and sex vulnerability to thalamic dysfunction after concussion

Recent studies have indicated the importance of age and sex vulnerability in the context of mTBI because older people and women are particularly predisposed to persistent PCS (Bazarian et al., 2010; Chen et al., 2016; Levin et al., 2021). This is also true for our data, where the symptom-resolved and symptom-prolonged groups demonstrated differences in terms of age and sex (Table 2)—possibly connoting the age and sexual vulnerability factor related to acute peri-thalamic injury. However, the predominance of thalamocortical coherence in the symptom-prolonged group can be explained by not only severe thalamic dysfunction but also age-related changes in the hemodynamic response shape, which slows down the neurovascular-coupled fMRI signals (Chen et al., 2015; West et al., 2019), particularly in older people with cerebral vascular diseases (Carusone et al., 2002; Li et al., 2021). Further investigation corroborating these findings and identifying the effects of age and sex on thalamic dysfunction are warranted.

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CRedit authorship contribution statement

Yi-Tien Li contributed to Conceptualization, Data curation, Formal analysis, computational modeling, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Project administration. Duen-Pang Kuo was involved with Data curation (rodent), and Formal analysis (rodent). Philip Tseng, Changwei W. Wu, Hsiao-Wen Chung and Yvonne W. Lui helped with Writing – review & editing. Yung-Chieh Chen, Sho-Jen Cheng and Li-Chun Hsieh contributed to Data curation (patient). Yung-Hsiao Chiang involved with Conceptualization, Project administration and Supervision of the research. Cheng-Yu Chen contributed to Conceptualization, Writing – review & editing, Project administration and Supervision of the research.

Declaration of Competing Interest

The authors declare that they have no competing interests.

Data Availability

Data will be made available on request. Open-source software is available from the resources as cited or from the authors upon request. Other source codes generated in this study can be downloaded at: <https://github.com/YiTienLi/matlab-toolbox>. The human data are not publicly available due to the nondisclosure agreement signed by the participants for their privacy. The translational rodent research data underlying this study are available at: https://drive.google.com/drive/folders/17DFyFz_OzyvpvB2Bx-jBOxgr3f3nvVIC9?usp=sharing.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Microsoft Bing in order to search for appropriate references. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.pneurobio.2023.102464](https://doi.org/10.1016/j.pneurobio.2023.102464).

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